

Trace Tables

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can trace variable values before and after each loop iteration.

I can use the excluded stop value in range correctly.

I can explain initialization and test zero one and several iterations.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

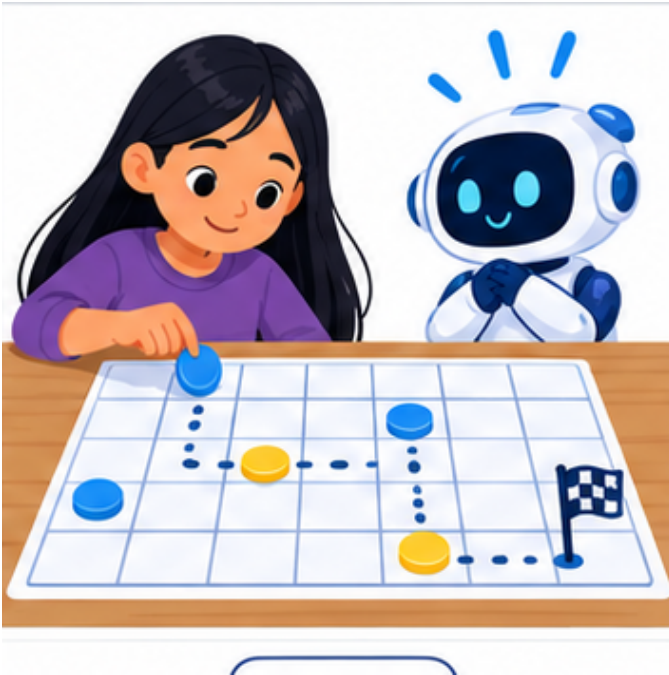
Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 4: Trace Tables

A trace table records changing state and reveals reasoning errors.



THINKER MISSION

Trace variables through a score-based game helper.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Starting with $\text{total} = 2$, what does $\text{total} = \text{total} + 3$ store?

This week's end goal

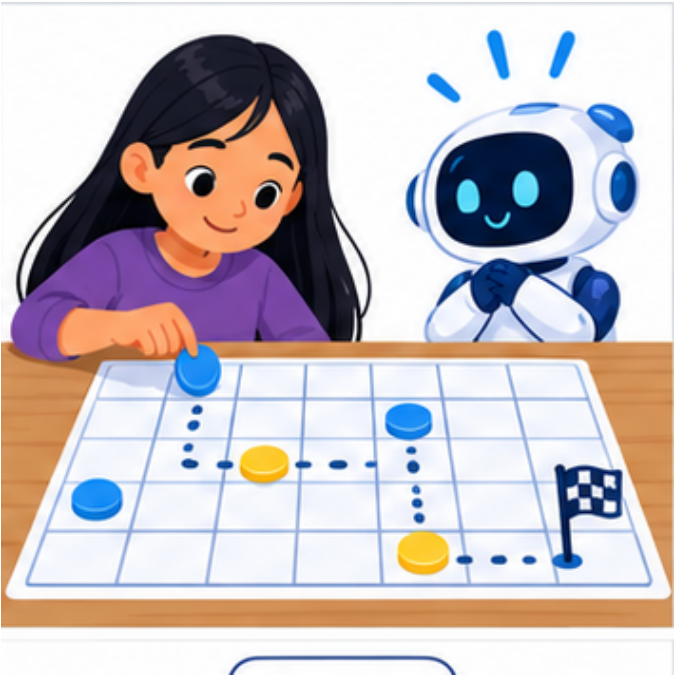
By the end, I can make a trace table for a for-loop, including zero iterations.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use trace tables to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Imagine adding numbered counters: first 1, then 2, then 3. A loop can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script. Tracing repeated calculations prepares you to inspect the data processing used in larger AI applications.

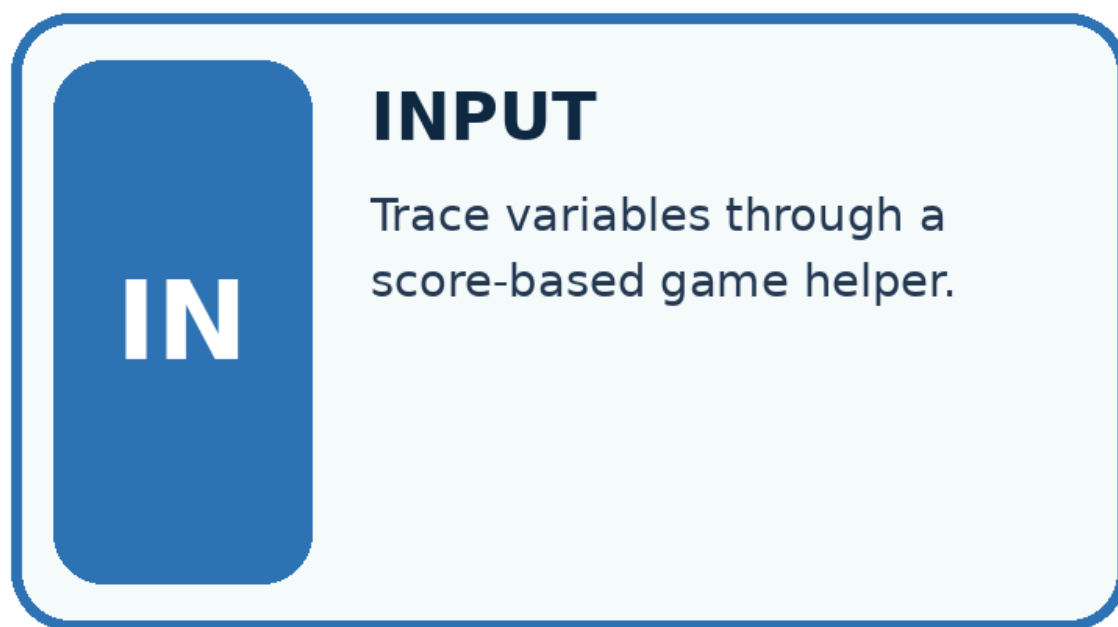
In for turn in `range(1, 4)`, Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body. This lesson uses integer range values with the default positive step of 1.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

A trace table records changing state and reveals reasoning errors.



OUT

OUTPUT

Trace variables through a score-based game helper.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
total = 0

for turn in range(1, 4):
    total = total + turn

print(total)
```

State after each addition

turn	total before	total after
------	--------------	-------------

turn: 1		
---------	--	--

total before: 0		
-----------------	--	--

total after: 1		
----------------	--	--

turn: 2		
---------	--	--

total before: 1		
-----------------	--	--

total after: 3		
----------------	--	--

turn: 3		
---------	--	--

total before: 3		
-----------------	--	--

total after: 6		
----------------	--	--

Initialize total to 0 before the loop. The unindented print runs once afterward.

The output is 6. Before the first iteration, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the loop. The table records state; it does not create extra instructions.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



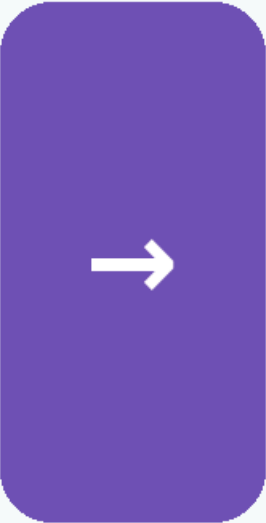
BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Use `range(0)` instead of `range(4)`.

Prediction: The loop body never runs and the printed total is 1.

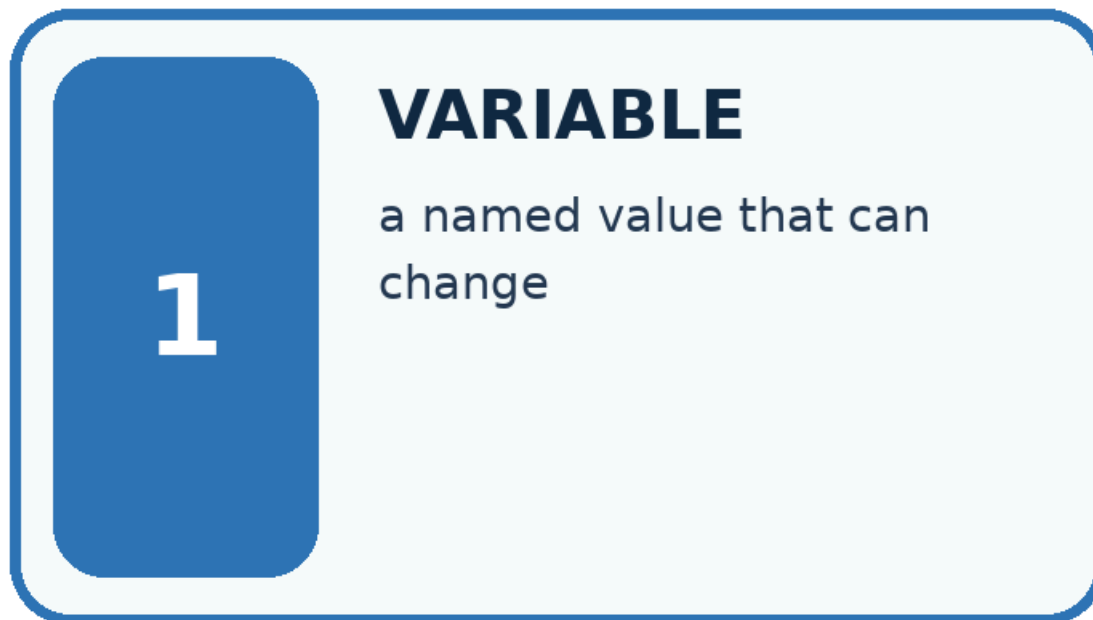
Reason: The iterable contains no turn values; the initial assignment still happened.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

STATE

all relevant values at one moment

3

TRACE

record each step in order

4

ITERATION

one pass through repeated steps

Word · My own example · Not this / boundary

Word: variable

My own example: A named reference to a value that can be used or changed in a program.

Not this / boundary: _____

Word: state

My own example: The current values and information that determine what a process does next.

Not this / boundary: _____

Word: trace

My own example: A record of the successive states, decisions or information transformations in a process.

Not this / boundary: _____

Word: iteration

My own example: One repeat of a process, often followed by inspection and revision.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Loop

Meaning: A block of instructions that can run repeatedly

Word: Accumulator

Meaning: A variable that keeps a running result

Word: Initialization

Meaning: Giving a variable its starting value

Word: Trace table

Meaning: A record of variable values as instructions execute

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 4 • WORKED EVIDENCE

- 1 Start score = 2.
- 2 Correct answer: $\text{score} = \text{score} + 3 \rightarrow 5$.
- 3 Hint used: $\text{score} = \text{score} - 1 \rightarrow 4$.
- 4 A table shows exactly where an unexpected value first appears.

1. Start score = 2.
2. Correct answer: score = score + 3 → 5.
3. Hint used: score = score – 1 → 4.
4. A table shows exactly where an unexpected value first appears.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Range expression · Values supplied · Number of iterations

Range expression: range(1, 4)

Values supplied: 1, 2, 3

Number of iterations: 3

Range expression: range(1, 2)

Values supplied: 1

Number of iterations: 1

Range expression: range(1, 1)

Values supplied: No values

Number of iterations: 0

Range expression: range(4)

Values supplied: 0, 1, 2, 3

Number of iterations: 4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0. Do not rely on a variable first assigned inside a loop: a zero-iteration run might never assign it.

Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Trace Tables

A fictional reading-token jar starts with one token and gains three per round.

The facts we will use

Python: total=1; for turn in range(4): total=total+3; after the loop print(total).

The update is inside the loop; print is outside.

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Trace Tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial state

Before the loop, total=1.



2. Iteration values

range(4) supplies turn values 0,1,2,3.



3. Updated totals

After each turn total is 4,7,10,13.



4. After the loop

Print 13 once, after all four updates.

CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Initial state

What happens and why: Before the loop, total=1.

Step: 2. Iteration values

What happens and why: range(4) supplies turn values 0,1,2,3.

Step: 3. Updated totals

What happens and why: After each turn total is 4,7,10,13.

Step: 4. After the loop

What happens and why: Print 13 once, after all four updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The turn value labels an iteration; it is not the number added. The fixed update adds three each time. Four iterations take the total from 1 to 13. Keeping print outside the loop produces one final display rather than four intermediate displays. Initialise total before the loop so a zero-iteration run still has a defined value.

What this example does not establish

This finite loop does not establish behavior of a while-loop with a different update or stopping condition.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A trace table records only the final answer. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Write the expected range values and trace table before running. Keep predicted and observed output separate, then explain the first row where they differ. For a loop that should add a fixed amount repeatedly, zero iterations should preserve the initial amount. One iteration checks one update; several check accumulation.

A learner checks the reset bug using only `range(1, 2)` and sees the expected answer 1. Explain why this does not establish correctness, and give a revealing test with its expected correct output.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET `evidence_record` PREDICT `expected_result` RUN one normal case and one boundary case
COMPARE `actual_result` with `expected_result` IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
total = 0

for turn in range(1, 4):
    total = 0
    total = total + turn

print(total)
```

This faulty version prints 3. The extra reset discards earlier additions on every iteration. Remove the indented reset while keeping initialization before the loop. Then test zero, one and several iterations: the one-iteration case alone cannot reveal this reset bug.

Moving `print(total)` into the correct loop produces three lines, 1 then 3 then 6. Moving it out produces only the final 6. Trace both values and output events. A loop follows written instructions; repeating them does not make the program learn.

Practise with less help: Trace before running

Use workbook W2 for the connected example “Loops and trace tables”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Move `print` inside the original loop. What sequence is displayed?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Use columns for step, event, old score, new score, and branch. Trace once by hand, predict the next row, then compare.

Use columns for step, event, old score, new score, and branch. Trace once by hand, predict the next row, then compare.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Trace each book's genre, length, difficulty, and total score before sorting.

THIS WEEK'S CHECKPOINT Trace each book's genre, length, difficulty, and total score before sorting.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Trace recommender loops with current item, eligibility, score and current best so each update can be

explained.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Trace variables through a score-based game helper.

CHALLENGE Trace variables through a score-based game helper.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Set score=2; add 3; if score>4 subtract 1; double score.

The question

Trace every value.

Worked explanation

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep the state after each instruction

Evidence or stage · Value or result

Evidence or stage: Set score

Value or result: 2

Evidence or stage: Add 3

Value or result: 5

Evidence or stage: If greater than 4, subtract 1

Value or result: 4

Evidence or stage: Double

Value or result: 8

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

2,5,4,8. The comparison is evaluated at 5 before subtraction.

14. Try a new case independently

Independent case: Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Trace each book's genre, length, difficulty, and total score before sorting.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.